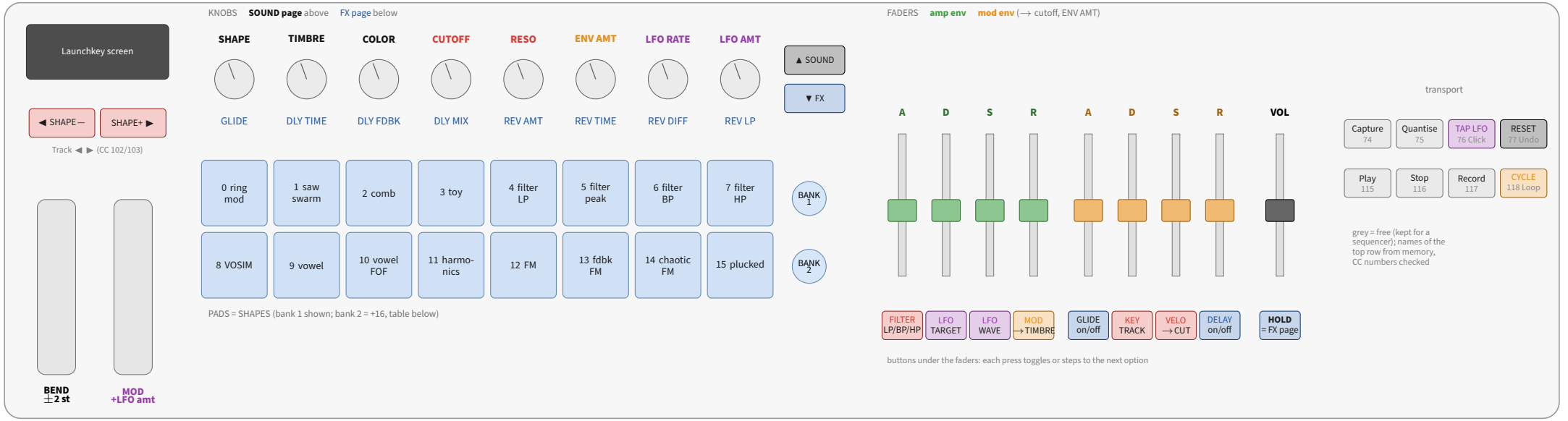
freak Axoloti Core + Launchkey 49 MK3 (standalone mode) · 3-voice Braids synth · patch `patches/freak/freak_v1.axp`



Knobs: two pages, scaled pickup

SHAPE	34 Braids shapes in turn (3–4 steps each)
TIMBRE / COLOR	the two Braids parameters (meaning depends on the shape)
CUTOFF	20 Hz $\cdot 2^{10k}$: 20 Hz – 20 kHz
RESO	Q = 0.5 – 20 (exponential)
ENV AMT	mod env → cutoff, ±5 octaves, 0 at centre
LFO RATE	0.05 – 20 Hz (or tap tempo)
LFO AMT	0 – 1, plus the mod wheel
GLIDE	0 – 1 s (quadratic), with GLIDE on
DLY TIME / FDBK / MIX	10 ms – 1.5 s; 0 – 0.95; 0 – 1
REV AMT/TIME/DIFF/LP	Elements reverb, 0 – 1 each

FX page: hold fader button 9 and turn, like a Shift key; let go and you are back on SOUND. To stay on FX hands-free, ▼ latches it and ▲ returns to SOUND. After a page change or RESET, a knob that is not where its value is still works: the value moves *proportionally* the way you turn and meets the knob at an end of travel at the latest. No jumps, no dead knobs. FX knobs 1–4 are silent until GLIDE or DELAY is on.

Envelopes and filter

Faders 1–4 **amp**, 5–8 **mod**: A, D, R = 1 ms $\cdot 10^{4k}$ (1 ms – 10 s, a decade per quarter of the travel); S = level 0 – 1. Exponential segments; attack aims at 1.2: a concave rise, fast at first (analog style).

Mod env → cutoff (ENV AMT), and → timbre (±0.5) with MOD→TIMBRE. **CYCLE** (Loop): it loops attack/decay while a key is held.

Filter: 12 dB/oct state-variable (LP/BP/HP, BP normalised). KEY TRACK (on at power-on): cutoff follows the keys, 1 oct/oct around note 60 (middle C). VELO→CUT: up to +2 octaves for a hard hit.

LFO (one, shared by the voices)

TARGET: **timbre** → color → cutoff (±3 oct) → pitch (±2 st). WAVE: **sine** → triangle → saw → square → S&H. TAP LFO: two taps 0.1–4 s apart set the period; LFO RATE takes over again when turned.

Pads: shapes, two banks

Bank 1 (0–15)
0 ring mod 1 saw swarm
2 comb 3 toy
4–7 filter LP/peak/BP/HP
8 VOSIM 9 vowel
10 vowel FOF 11 harmonics
12 FM 13 feedback FM
14 chaotic FM 15 plucked

Bank 2 (16–31)
16 bowed 17 blown
18 fluted 19 bell
20 drum 21 kick
22 cymbal 23 snare
24 wavetables 25 wave map
26 wave line 27 paraphonic
28 filtered noise 29 twin peaks
30 clocked noise 31 granular

Top row = first 8, bottom row = next 8. 32 particle noise and 33 digital modulation: ▶ from 31, or the SHAPE knob. Strike-type shapes (plucked, bell, drums) are struck on each key press.

Notes

3 voices (a 4th note takes the oldest voice). Power-on: saw swarm, LP, key tracking on, delay off. RESET (Undo) restores that sound. The toggles' states are **not shown** yet (no LEDs in standalone mode). Each press moves to the next option, so count, or listen. Page 2: shapes, dial values, starting points.

The Launchkey's own features

They make notes, so they play freak as well (set them on the Launchkey, see its user guide):

Arp: up, down, up/down, as played, random, chord, mutate; 1/4 – 1/32T; 1–4 oct; latch, gate, swing, rhythm.
Scale: minor, major, **Dorian**, **Mixolydian**, Phrygian, harmonic minor, minor/major pentatonic.
Fixed chord, octave ±, transpose.
Not yet tested with freak.

Signal path and load

3 × (Braids → SVF → amp env × velocity) → delay → Elements reverb → VOL. Velocity: soft notes keep $\frac{1}{4}$ of the level. DSP 54.5 %, patch RAM 43 of 44 KB (Axoloti Core, Ksoloti 1.1.0).

The 34 shapes: what TIMBRE (knob 2) and COLOR (knob 3) do (after the Braids manual)

#	pad	shape	TIMBRE	COLOR
0	1·T1	**ring mod**	tuning of sines 2, 3	tuning of sines 2, 3
1	1·T2	**saw swarm**	detune (7 saws)	high-pass
2	1·T3	**comb**	comb pitch	feedback — / 0 / + (L / noon / R)
3	1·T4	**toy**	clock rate	glitches
4	1·T5	**filter LP**	cutoff	wave: saw→square→tri
5	1·T6	**filter peak**	cutoff	wave: saw→square→tri
6	1·T7	**filter BP**	cutoff	wave: saw→square→tri
7	1·T8	**filter HP**	cutoff	wave: saw→square→tri
8	1·B1	**VOSIM**	formant 1	formant 2
9	1·B2	**vowel**	vowel a e i o u	formant shift
10	1·B3	**vowel FOF**	vowel a e i o u	formant shift
11	1·B4	**harmonics**	centre of the 12 harmonics	their spread
12	1·B5	**FM**	mod amount	ratio
13	1·B6	**feedback FM**	mod amount (harsher)	ratio
14	1·B7	**chaotic FM**	mod amount (unstable)	ratio
15	1·B8	**plucked**	damping	pluck position
16	2·T1	**bowed**	friction	bow position
17	2·T2	**blown**	air pressure	geometry
18	2·T3	**fluted**	air pressure	geometry
19	2·T4	**bell**	damping	inharmonicity
20	2·T5	**drum**	damping	brightness
21	2·T6	**kick**	decay	tone
22	2·T7	**cymbal**	band-pass cutoff	drone ↔ noise
23	2·T8	**snare**	tone	snappy (noise)
24	2·B1	**wavetables**	sweep the table	table, 1 of 20
25	2·B2	**wave map**	X in a 16×16 map	Y
26	2·B3	**wave line**	position	smoothing (left: stepped)
27	2·B4	**paraphonic**	wave	**chord** (left: unison)
28	2·B5	**filtered noise**	resonance	LP ↔ HP; keys = cutoff
29	2·B6	**twin peaks**	resonance Q	peak spacing
30	2·B7	**clocked noise**	periodicity	quantise 2–32 steps
31	2·B8	**granular**	grain density	pitch randomness
32	▶	**particles**	density	pitch randomness
33	▶	**digital mod**	bit rate	data byte

pad = bank·row+column: 1·T1 is bank 1, top row, first pad (BANK 1/2 = the round buttons right of the pads). Plucked, bowed, bell, drum, kick, cymbal and snare are struck on each new key: S and R shape the tail only. Time constants inside Braids that do not follow pitch (grain rates, built-in decays) run at half the original speed.

Starting points (not yet tried by ear: adjust to taste; fractions = how far a knob/fader is turned)

Vowel pad (nu-jazz)	shape 10 vowel FOF; TIMBRE 1/2, COLOR 1/2; LFO→timbre, sine, RATE 1/4 (0.2 Hz), AMT 1/3; amp A 2/3, S 3/4, R 3/4; CUTOFF 3/4; FX: DELAY on, TIME 1/2, FDBK 1/3, MIX 1/4; REV AMT 1/2.
FM bass (broken beat)	shape 12 FM, play an octave down; TIMBRE 1/4, COLOR 1/2; amp A 0, D 1/2, S 0, R 1/4; CUTOFF 1/3, RESO 1/4, ENV AMT 3/4, mod D 1/2; VELO→CUT on; GLIDE on with GLIDE 1/4 for slides.
Glass mallets	shape 19 bell; TIMBRE 1/3, COLOR 1/4; amp S full, R 3/4; filter HP (press FILTER twice), CUTOFF 1/4; REV AMT 3/4, TIME 3/4.
Chord stabs	shape 27 paraphonic: every key plays a chord, COLOR picks it; Launchkey Scale mode Dorian keeps it modal; amp D 1/2, S 0; DELAY on, TIME 3/4, FDBK 1/2 for dub echoes.
Granular weather	shape 31 granular; TIMBRE 1/2, COLOR 3/4; LFO→color (TARGET once), S&H (WAVE 4 times), tap the tempo; CYCLE (Loop) on with ENV AMT 3/4, filter BP: rhythmic swells.
Pitched rain	shape 32 particles (▶ from 31); TIMBRE 1/4, COLOR full; DELAY on, FDBK 3/4; REV AMT 3/4.

Where the values fall (knob or fader turned . . .)

	1/4	1/2	3/4	full
A, **D**, **R** (both envelopes)	10 ms	100 ms	1 s	10 s
CUTOFF (at note 60)	113 Hz	640 Hz	3.6 kHz	20 kHz
RESO (Q)	1.3	3.2	8	20
ENV AMT	−2.5 oct	0	+2.5 oct	+5 oct
LFO RATE	0.22 Hz	1 Hz	4.5 Hz	20 Hz
GLIDE	0.06 s	0.25 s	0.56 s	1 s
DLY TIME	35 ms	0.12 s	0.43 s	1.5 s
DLY FDBK	0.24	0.48	0.71	0.95
VOL (gain)	0.06	0.25	0.56	1

Times are exponential: each quarter of the travel is a factor of 10. S, TIMBRE, COLOR and the mixes are linear, 0–1.

Something seems wrong?

No sound: VOL (fader 9) up? CUTOFF (knob 4) not at the bottom in LP, nor at the top in HP? A struck shape (pluck, bell, drums) only sounds on a new key. Is the patch running (Live / startup)?

Knobs do nothing: are you holding button 9 (FX page)? Did ▼ latch FX? Press ▲. FX knobs 1–4 need GLIDE / DELAY on.

Wobbles or detunes: LFO AMT (knob 8) or the mod wheel; TARGET may be on pitch.

A note drops out: 3 voices; a 4th note takes the oldest.

Echoes pile up: lower DLY FDBK or switch DELAY off (the tail dies away).

Lost: RESET (Undo) restores the power-on sound.

With the patcher connected, its log prints page changes, resets, knobs catching up and unknown messages.

MIDI reference (Launchkey 49 MK3, standalone)

keys, bend, mod wheel	ch 1: notes 36–84, pitch bend, CC 1
knobs 1–8	ch 1 CC 21–28
faders 1–9	ch 1 CC 71–79
fader buttons 1–9	ch 1 CC 11–19 (127 press, 0 release)
pads	ch 10 notes 36–51
arrows ▲/▼, track ◀/▶	ch 16 CC 51/52, 102/103
round buttons, Click, Undo, Loop	ch 16 CC 104/105, 76, 77, 118

Recorded from this unit (`docs/launchkey49mk3_map.md`).

Jazz harmony on 3 voices

Shell voicings: let the 3 voices carry the colour, not the root: 3rd, 7th and one tension. In C: Cmaj9 = E–B–D, Dm9 = F–C–E, G13 = F–B–E, G7♭9 = F–B–A♭. A bass line from another instrument supplies the roots.

More notes per key: shape 27 (paraphonic) plays a chord on every voice (COLOR picks it), and the Launchkey's **Fixed Chord** repeats a stored shape from one key. **Scale** mode Dorian or Mixolydian keeps every key in the mode.